VEHICLE FORGE

Companion Guide

Version 2.1 — Glassforge Games Ltd.

A vehicle construction toolkit for Savage Worlds

Six sliders. Unlimited vehicles. Under a minute.

glassforge.pages.dev

What’s New in v2.1

Version 2.1 is a complete overhaul of the vehicle construction interface. Seventeen separate controls — buttons, selectors, and modifications — have been replaced with six unified sliders. The result is a tool that builds combat-ready vehicles in under a minute while maintaining 94% accuracy against 102 published Savage Worlds vehicles.

The Six Sliders

Every vehicle stat is now controlled by a slider with a descriptive label. Size sets the vehicle’s scale. Toughness and Armour define its durability. Speed, Handling, and Wounds complete the picture. No more digging through modification catalogues for Improved Speed or calculating frame type multipliers. Just slide and go.

Removed in v2.0–2.1

Armour Technology buttons (Primitive, Forged, Industrial, Composite, Nanoweave, Exotic) — replaced by the Armour slider. Frame Type selector (Civilian, Rugged, Military, Futuristic) — absorbed into cost scaling. Hull Plating, Composite Layering, Reinforced Frame, Weak, Tough, Improved Speed, Underpowered, High-Performance Engine, Improved Handling, Sluggish, and Weak Frame modifications — all absorbed into the five stat sliders. Base stats info box — redundant with slider display.

Added in v2.0–2.1

Toughness slider with domain-aware baselines, Armour slider with domain-aware baselines, Speed slider, Handling slider, Wounds slider, overflow mechanic for extreme builds, weapon grade recalculation on linking, audio feedback on all interactions, compact slider layout, three-column locomotion grid.

Quick Start

Building a vehicle takes four steps and under a minute:

Step 1 — Set the Size slider to match your vehicle’s scale. The description line shows length, weight, and real-world examples. A motorcycle is Size 1, a car is Size 4, a tank is Size 8–9, a destroyer is Size 17–18.

Step 2 — Pick a locomotion type. Wheeled, tracked, hover, aircraft (VTOL, turboprop, jet), watercraft (sail, propeller, jet), or walker (bipedal, multileg). This sets the base Speed and determines armour scaling.

Step 3 — Adjust the five stat sliders. Toughness, Armour, Speed, Handling, and Wounds each start at Standard. Slide left for lighter or weaker, right for tougher or faster. The labels tell you what you’re getting — Fragile, Lightweight, Standard, Robust, Ironclad.

Step 4 — Add modifications and weapons. Drag from the catalogue to your vehicle. Pick mount types (Pintle, Fixed, Turret). Link weapons for extra firepower. The stat block generates automatically.

That’s it. Save, export, print, or batch-export your entire garage.

The Slider System

Six sliders control everything about a vehicle. Each has a descriptive label, overflow buttons for extreme builds, and a base value derived from Size or locomotion. All sliders sit together in the Vehicle Stats panel.

Slider	Controls	Base From	Step	Range
Size	Scale, crew, mod slots, cost	Direct value	1	1–30
Toughness	Structural mass (not reduced by AP)	Size + 5	Size-dependent	±5
Armour	Protective plating (reduced by AP)	Domain-aware	Domain-aware	±5
Speed	Movement rate	Locomotion type	±1	±5
Handling	Manoeuvrability	Size category	±1	±5
Wounds	Damage capacity	Size category	±1	−3 to +2

The overflow buttons (+ and −) extend beyond the slider range, following the d12+X convention. A vehicle at Maximum +3 has three clicks of extra armour beyond the slider cap. 94% of canon vehicles fit within the standard ranges — overflow is for edge cases.

Size

Size (1–30) determines the vehicle's scale. Everything else derives from it: base Toughness, Wounds, Handling, crew capacity, mod slots, and cost. The description line shows approximate length, weight, and real-world examples.

Base Cost: 1,000 × Size²

Mod Slots: max(3, floor(Size × 2.5))

Size Categories

Category	Size	Handling	Wounds	Crew	Examples
Light	1–3	+1	3	1	Motorcycle, drone, chariot, light walker
Medium	4–7	0	4	1	Car, jeep, APC, elephant, light tank
Heavy	8–11	−1	5	5	MBT, heavy bomber, corvette, locomotive
Super-Heavy	12–20	−2	6	20	Frigate, destroyer, carrier, fortress
Capital	21–25	−3	7	100	Mobile fortress, mega-carrier, space station
Titan	26+	−4	8	500	World-engine, orbital platform, generation ship

Sizes 21+ are speculative and intended for science fiction or high fantasy campaigns. Sizes 26–30 use miles rather than feet for length descriptions.

Toughness Slider

The Toughness slider controls structural integrity — the vehicle's inherent mass, frame strength, and construction quality. This is the portion of total Toughness that Armour Piercing cannot reduce.

Base Value: Size + 5 (universal across all domains)

Step Size: max(1, round(Size / 4) + 1) — larger vehicles have bigger steps

Structural Toughness: Base + (slider position × step)

Toughness Labels

Position	Label	Description
−5	Fragile	Extremely fragile construction for this class.
−4	Flimsy	Very flimsy. Will not survive serious impacts.
−3	Lightweight	Lightweight construction. Sacrifices durability for speed.
−2	Light	Lighter than average for this size class.
−1	Slightly Light	Slightly below standard structural strength.
0	Standard	Standard structural strength for this size class.
+1	Sturdy	Sturdier than typical. Reinforced frame or heavier materials.
+2	Robust	Robust construction. Built to take punishment.
+3	Very Tough	Very tough. Significantly over-engineered.
+4	Extremely Tough	Among the most solid of this size.
+5	Ironclad	Maximum structural integrity for this class.

Step Size Table

Size	Step	Min Struct	Std Struct	Max Struct
0–3	1	0	5–8	10–13
4–7	2	-1–1	9–12	19–22
8–11	3	-2–2	13–16	28–31
12–15	4	-3–1	17–20	37–40
16–19	5	-4–0	21–24	46–49
20+	6+	varies	Size+5	varies

Armour Slider

The Armour slider controls protective plating, shielding, or whatever defensive layer the GM imagines for their setting. This is the parenthetical value in the Toughness stat — the number that Armour Piercing specifically targets.

Base Value: Domain-aware — varies by locomotion group and Size

Step Size: Domain-aware — varies by locomotion group and Size

Armour Value: Base + (slider position × step)

Total Toughness: Structural Toughness + Armour Value

The slider is setting-agnostic. No materials are named. “Maximum” might mean dragon-scale plate in a fantasy game, composite ceramics in a modern setting, or energy shielding in science fiction. The GM provides the fiction.

Armour Labels

Position	Label	Description
−5	Unprotected	Virtually no protection. Civilian bodywork, canvas, or thin hull.
−4	Minimal	Negligible protection. Light commercial construction.
−3	Light	Light protection. Hardened or lightly armoured.
−2	Below Average	Below average for a combat vehicle. Scout or recon class.
−1	Modest	Modest protection. Light military or up-armoured civilian.
0	Standard	Standard military protection for this size class.
+1	Reinforced	Better protected than typical military vehicles.
+2	Heavy	Heavy protection. Elite or next-generation military.
+3	Superior	Superior protection. Cutting-edge or experimental.
+4	Extreme	Among the toughest of this size class.
+5	Maximum	Pushing the limits of known engineering.

Domain-Aware Baselines

When you select a locomotion type, the armour base and step automatically adjust. Ground vehicles and walkers share one scale. Watercraft use a different scale that accounts for the jump from patrol boats to belt-armoured warships. Aircraft use a lighter scale reflecting the weight-sensitivity of flight.

Ground / Walker Armour

Size	Base	Step	At −5	At 0	At +5
0	2	1	−3	2	7
3	3	1	−2	3	8
4	4	2	−6	4	14
6	12	2	2	12	22
8	16	3	1	16	31
10	20	4	0	20	40
12	24	5	−1	24	49
14	28	6	−2	28	58

Water Armour

Size	Base	Step	Notes
0–11	1–5	sz/3	Small craft, patrol boats
12–16	6–8	4	Corvettes through frigates
17+	18–46	4	Destroyers through carriers

Belt-armoured warships (destroyers, battleships) sit at Standard or above. An SFC-class destroyer might need Maximum +3 to reach its canonical armour value — that's what the overflow buttons are for.

Speed, Handling & Wounds

These three sliders use simple ±1 steps with no domain scaling. The base values come directly from the locomotion type (Speed) or Size category (Handling and Wounds).

Speed

Base speed is set by locomotion type. Each slider position adds or subtracts 1. Labels: Crawling, Very Slow, Slow, Below Average, Slightly Slow, Standard, Quick, Fast, Very Fast, Blazing, Maximum.

Locomotion	Base Spd	Approximate MPH at Standard
Wheeled	4	30–40 MPH
Tracked	3	20–30 MPH
Hover	5	40–50 MPH
VTOL	7	100–140 MPH
Turboprop	10	250–400 MPH
Jet	12	500+ MPH
Sail	2	5–10 MPH
Water Prop	3	15–25 MPH
Water Jet	4	30–45 MPH
Biped Walker	6	30–40 MPH
Multileg Walker	5	25–35 MPH

Speed Labels

Position	Label
−5	Crawling
−4	Very Slow
−3	Slow
−2	Below Average
−1	Slightly Slow
0	Standard
+1	Quick
+2	Fast
+3	Very Fast
+4	Blazing
+5	Maximum

Handling

Base handling comes from the Size category. Each slider position adds or subtracts 1. Labels: Unwieldy, Very Clumsy, Clumsy, Awkward, Slightly Stiff, Standard, Responsive, Agile, Very Agile, Nimble, Razor-Sharp.

Wounds

Base wounds come from the Size category. The slider range is tighter (−3 to +2) because canon vehicles rarely deviate from the Size-based default. The overflow buttons allow further adjustment for unusual builds. Labels: Fragile, Brittle, Delicate, Standard, Resilient, Rugged.

The Overflow Mechanic

Most vehicles fit comfortably within the −5 to +5 range. But some extreme builds push beyond — a heavily armoured battleship, a future super-tank, a dwarven ironclad with impossibly thick plating. The + and − buttons extend the slider range indefinitely, following the d12+X convention.

Clicking + when the slider is already at +5 moves the value to +6, displayed as “Maximum +1”. Continue clicking for +7 (“Maximum +2”), +8 (“Maximum +3”), and so on. The same works in the negative direction: below −5 shows “Unprotected −1”, “Unprotected −2”, etc.

Canon Vehicles Requiring Overflow

Vehicle	Size	Armour Position	Label
SFC Destroyer	12	+8	Maximum +3
SFC Battleship	16	+10	Maximum +5
Future Hover Car	3	+6	Maximum +1
Future APC	6	+6	Maximum +1
Future Hover Loco	11	+6	Maximum +1

Only 6 out of 102 canon vehicles need overflow. The slider handles 94% of builds without touching the buttons.

Glassforge Threat Grades

The Threat Grade system lets GMs match weapon scale to vehicle durability at a glance. Every vehicle has a Defence Grade based on total Toughness. Every weapon has a Weapon Grade based on average damage plus AP. Both use the same A–F scale, so the comparison is instant: a Grade C weapon is a serious threat to a Grade C vehicle.

Defence Grade

Grade	Total Toughness	Typical Examples
A	0–15	Motorcycles, civilian cars, light drones
B	16–25	Military jeeps, APCs, patrol boats
C	26–35	Light tanks, armoured vehicles
D	36–45	Main battle tanks, heavy AFVs
E	46–55	Future MBTs, heavy warships
F	56+	Super-heavy tanks, capital ships

Each grade has +/− subdivisions based on position within the band. A vehicle near the top of B band shows B+, near the bottom shows B−. This is Glassforge’s own IP-clean rating system, distinct from any published source.

Weapon Grade

Calculated from average damage plus Armour Piercing. Linked weapons recalculate: dual-linked adds +2 damage, quad-linked adds +4 damage to the threat score. A base Grade B− weapon can become a solid B or B+ when quad-linked. The catalogue shows base grades. The hardpoints bay shows adjusted grades.

Weapons & Linking

Drag weapons from the catalogue to the hardpoints bay. Click the mount badge to cycle between Pintle, Fixed Front, and Turret. Click Link to cycle through unlinked, dual-linked (×2), and quad-linked (×4).

Mount Types

Mount	Arc	Mod Cost	Price
Pintle	180° swivel	Base	Base
Fixed Front	45° forward	Half	Base
Turret	360° rotation	Double	Double

Linked Weapons

Config	Shooting	Damage	Extra Mods	Cost
Single	—	—	—	×1
Dual Linked	+1	+2	+1	×2
Quad Linked	+2	+4	+3	×4

Era Filtering

The weapon catalogue filters by tech era: Ancient, Medieval, Blackpowder, Industrial (WW1/WW2), Modern, Future (Sci-Fi), and Advanced (Far-Future). This keeps the catalogue focused on weapons appropriate to your setting.

Cost Model

Base Cost: 1,000 × Size²

Slider Multiplier: Combined absolute slider positions scale cost progressively

Modifications: Each mod adds cost based on Size

Weapons: Base cost, doubled for turret mounts, multiplied by link count

The cost model is intentionally approximate. Vehicle prices in published Savage Worlds products vary enormously based on setting, era, and narrative context. The model gives GMs a reasonable starting point for comparing builds rather than an authoritative price list.

Features

Save & Load: Vehicles stored in browser localStorage. Unlimited saves per browser

Export/Import: JSON format (.cvf.json) for sharing and backup

Batch Export: Export all saved vehicles to a single file for library distribution

Print View: Clean printable stat blocks with vehicle image

Vehicle Image: Upload or paste an image. Appears in stat block output

Extensions: Import custom weapon packs (.vfx files) for setting-specific arsenals

Undo/Redo: Full state undo across all changes

Era Filter: Filter weapons by tech era: Ancient through Far-Future

Audio Feedback: Click and clunk sounds on all interactions (toggleable in Settings)

Comparison View: Compare two saved vehicles side-by-side

Reference Guide: Built-in weapons table, Savage Worlds chase rules, and grade system documentation

Appendix A: Canon Validation

The slider system was calibrated against 102 canon vehicles from Savage Worlds core rules and the Vehicle Guide. The following table shows accuracy across vehicle categories.

Category	Canon Count	Within ±2 Pts	Accuracy
Civilian Cars	12	12	100%
WW2 Tanks	6	6	100%
Modern MBTs	4	3	75%
APCs/IFVs	4	4	100%
Future Vehicles	5	5	100%
Aircraft	12	11	92%
Watercraft	8	7	88%
TOTAL	102	96	94.1%

The 6% of vehicles outside ±2 points are primarily edge cases requiring overflow: heavily armoured warships and some future military vehicles with extreme protection ratings.

Appendix B: SWADE Core Rules Vehicles

All named vehicles from the Savage Worlds Adventure Edition core rulebook, showing how each maps to Vehicle Forge slider settings.

Civilian Ground Vehicles

Vehicle	Size	Loco	Tough	Hand	Spd
Carriage	3	Wheeled	6	T-2, A-3, H-3	S-2
Early Car	3	Wheeled	7	T-1, A-3, H-2	S-1
Street Bike	1	Wheeled	8	T+1, A-2	S+2
Dirt Bike	0	Wheeled	7	T+1, A-2	S 0
Compact Car	4	Wheeled	10(2)	H+1	S+1
Mid-Sized Car	4	Wheeled	11(2)	T+1	S+1
Minivan	4	Wheeled	12(2)	T+2	S 0
SUV	5	Wheeled	14(2)	T+2	S+1
Sports Car	4	Wheeled	10(2)	H+2	S+3
Semi-Truck	9	Wheeled	14(2)	T-1, A-2	S 0

WW2 Military Ground Vehicles

Vehicle	Size	Loco	Tough	Notes
Jeep	4	Wheeled	10(2)	HMG pintle, 4WD, H+1
M4 Sherman	8	Tracked	24(8)	Med tank gun turret, A-1
M5A1 Stuart	7	Tracked	21(7)	Light tank gun turret, H+1, A-1
T-34/76	7	Tracked	24(8)	Med tank gun turret, sloped
Panzer IVJ	7	Tracked	26(10)	Med tank gun turret, S-1, A+1
Tiger II	8	Tracked	34(16)	Heavy tank gun turret, T+1, A+2, H-1, S-1

Modern Military Ground Vehicles

Vehicle	Size	Tough	Notes
M1A1 Abrams	9	57(37)	Tank cannon turret, T+2, A+4, night vision
M2 Bradley	7	22(6)	Autocannon turret, H+1, night vision
T-72 MBT	9	43(25)	Tank cannon turret, T+1, A+2, S+1
T-80 MBT	8	52(32)	Tank cannon turret, T+2, A+3, night vision
BTR-70 APC	6	20(5)	HMGs turret, H-1, S+1, amphibious

Aircraft

Vehicle	Size	Loco	Tough	Notes
Biplane	4	Turboprop	12(1)	S-2, A-2
Cessna	5	Turboprop	12(2)	H+1, S-2
Helicopter	7	VTOL	12(2)	S-1, T-2, A-1
Learjet	8	Jet	16(2)	H+3, S-1, T-1, A-2
Space Shuttle	13	Jet	20(4)	H+1, S+5, T-2, A-2
B-17	10	Turboprop	19(2)	H-1, S-4, T-2, A-2, linked HMGs

BF-109	6	Turboprop	13(2)	H+1, S-1, cannon + linked HMGs
P-51 Mustang	6	Turboprop	13(2)	H+1, 3× linked HMGs
Spitfire	6	Turboprop	13(2)	H+2, S-1, linked HMGs + cannon

Appendix C: Vehicle Guide Conversion Reference

The Savage Worlds Vehicle Guide provides conversion tables, not named vehicles. These tables show typical Toughness values by Size for each domain and era. Vehicle Forge's slider system matches these baselines at Standard (slider position 0) and allows GMs to build anywhere in the range.

The Vehicle Guide covers: Civilian Ground Vehicles (Size −4 to 14), Modern Era Military Ground (Size −2 to 14), Future Era Military Ground (Size −3 to 14), Civilian Watercraft (Size −2 to 26), Military Watercraft (Size −2 to 25), Civilian Aircraft (Size −2 to 15), Military Aircraft (Size −3 to 15), and Spacecraft (Size 0 to 40).

For specific conversion examples, import the Canon Vehicles batch file (SWADE_Canon_Vehicles_Batch.cvf.json) which includes all named SWADE core rules vehicles pre-built.

Appendix D: Open Questions & Next Steps

Stat Block Polish

The Special line in the stat block currently includes full walker rules text. This should be condensed to keywords only (“Walker”) with full rules remaining in the Reference Guide tab.

UI Grid Standardisation

A 3×2 slider grid layout with an info panel is planned. All six sliders in three rows of two, with calculated values and brief guidance notes displayed alongside each pair.

Price Calibration

The cost multiplier is a placeholder. Canon vehicle prices vary enormously by setting and era. A calibration pass against published prices would improve accuracy for GMs who care about economic consistency.

Space & Airship Locomotion

Spacecraft and airship locomotion types are deferred to dedicated expansion supplements. The Vehicle Guide’s spacecraft tables extend to Size 40 with speeds measured in multiples of 20,000 MPH — a fundamentally different scale that deserves its own treatment.

Extension Packs

Custom weapon and vehicle packs (.vfx files) allow third-party content. Planned packs include Age of Sail Fleet, Viking Raiders, Napoleonic Navy, WW2 Theatre-specific vehicles, and more. See the Master Plan for the full catalogue.

Appendix E: Version History

Version	Date	Changes
v2.1	Feb 2026	Speed, Handling, Wounds sliders. Size moved to slider grid. 6 more mods removed. Base stats box removed. Compact slider layout. Loco grid 3-col. Weapon grades reflect linking. Audio on all actions. Size label renamed
v2.0	Feb 2026	Two-slider armour system. Removed AT buttons, 5 defensive mods, Frame Type system. Domain-aware baselines. Overflow mechanic. Display format fix
v1.7	Feb 2026	Armour Technology button fix, single source of truth for AT data
v1.6	Feb 2026	Toughness calc fix, clipboard export parity
v1.4	Jan 2026	Companion Guide first edition, Glassforge Threat Grade system
v1.0	Jan 2026	Major rebuild: drag-and-drop mods/weapons, colour-coded mounts, linked weapons, comparison view

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